

$\mathcal{O}(n)$ alternative to Quantum Fourier Transform with efficient neural net classical post-processing

Kaiming Bian,^{1,2,*} Zujin Wen,^{3,*} and Oscar Dahlsten³

¹*Shenzhen Institute for Quantum Science and Engineering,
Southern University of Science and Technology, Shenzhen, 518055, China*

²*International Quantum Academy, Shenzhen, 518048, China*

³*Department of Physics, City University of Hong Kong,
Tat Chee Avenue, Kowloon, Hong Kong SAR, China*

(Dated: July 10, 2026)

The Quantum Fourier Transform (QFT) is required by hidden subgroup problem (HSP) algorithms, including Shor’s algorithm for factoring. The circuit depth of the QFT remains challenging for near-term hardware. To find shallower alternatives we identify two properties that are exploited by the QFT to enable HSP. Firstly, the shift invariance of the QFT allows for the removal of a random overall shift. Secondly, the QFT retains information about the hidden subgroup generator accessible in the measurement outcomes. We quantify that information via the discrete Fisher information. We construct a family of shallow circuits using Hadamards and controlled-Phase gates, HP- L circuits, that we prove preserve shift invariance. Numerical analysis shows these circuits retain exponentially growing Fisher information. The $\mathcal{O}(n)$ HP-1 can replace the $\mathcal{O}(n^2)$ QFT in Shor’s algorithm, as demonstrated numerically, with an efficient neural network implementing classical post-processing.

Introduction.—The $\mathcal{O}(n^2)$ circuit depth required by the standard Quantum Fourier Transform (QFT) presents experimental challenges. The QFT is used in the exponential quantum speedups of landmark protocols—such as Shor’s algorithms for prime factorization and discrete logarithms [1–3]. The associated family of algorithms, all using the QFT, is known as the (quantum) hidden subgroup problem (HSP) algorithms [4–7]. The experimental implementation of the QFT remains challenging. For a 100-qubit system ($n = 100$), the QFT demands thousands of circuit layers, which vastly exceeds current hardware capabilities. For instance, IBM’s 127-qubit Eagle processor is limited to a depth of 60 layers even with extensive error mitigation [8]; Google’s 105-qubit Willow processor reaches its operational threshold at around 40 layers [9]; USTC’s 105-qubit Zuchongzhi 3.0 processor demonstrated random circuit sampling on 83-qubits, with around 32 layers [10]. Moreover, for quantum error-mitigation protocols that are commonly used, the sampling overhead can scale exponentially with the circuit depth [11, 12].

There has been significant progress in reducing the circuit depth required. These efforts have enabled record experimental QFT fidelities [13] through techniques including (i) approximate or truncated QFT constructions, which omit small-angle controlled rotations [14, 15] with $\mathcal{O}(n \ln n)$ depth, (ii) dynamic mid-circuit measurements, which reduce the resource scaling of QFT followed immediately by measurement from $\mathcal{O}(n^2)$ two-qubit gates to $\mathcal{O}(n)$ mid-circuit measurements without connectivity constraints [16], and (iii) quantum circuit cutting and cluster-simulation methods, which trade coherent quantum depth for additional sampling and classical post-processing overhead [17, 18].

We here take a different approach to reducing the circuit depth required. Rather than approximate the QFT we identify key features of the QFT that make it compatible with the HSP. We focus on shift-invariance combined with retention of information about the generator in question. Fig. 1(a) summarizes the main idea. We design non-QFT circuits that retain those key features.

We construct a circuit family, called HP- L circuits, that preserves shift invariance. In particular, the HP-1 circuits are shallow and retain information about the generator. The HP- L circuits interleave transversal Hadamard (H) gates with L layers of phase (P) blocks. We prove analytically that HP- L circuits are shift-invariant. Numerical scaling analysis demonstrates exponentially increasing retention of Fisher information $\sim e^{kn}$, albeit with a worse exponent k than QFT. That $\sim e^{kn}$ -scaling consistent with efficient scaling of the number of samples required.

We numerically simulate replacing the QFT in Shor’s algorithm with HP-1 circuits, and using an efficiently scaling neural network [19] for classical post-processing. Factorisation is successfully achieved.

Necessary condition for Shift Invariance.—The standard quantum HSP algorithm is illustrated in Fig. 1(b) with gray blocks [1–3, 5]. After U_f , the state is proportional to $\sum_x |x\rangle |f(x)\rangle$. The elements x belong to a finite abelian group G with group operation $+$. We are guaranteed that $f(x+r) = f(x)$ for the unknown generator r that we wish to find. In other words, the function $f(\cdot)$ is constant on cosets of V , the subgroup generated by r . Upon measuring (or imagining measuring) the second register to find $|f(c)\rangle$ the state becomes proportional to $\sum_{q \in \{0,1,2,\dots\}} |c+qr\rangle |f(c)\rangle$, where the shift c acts as a random coset offset and does not affect the recovery of r .

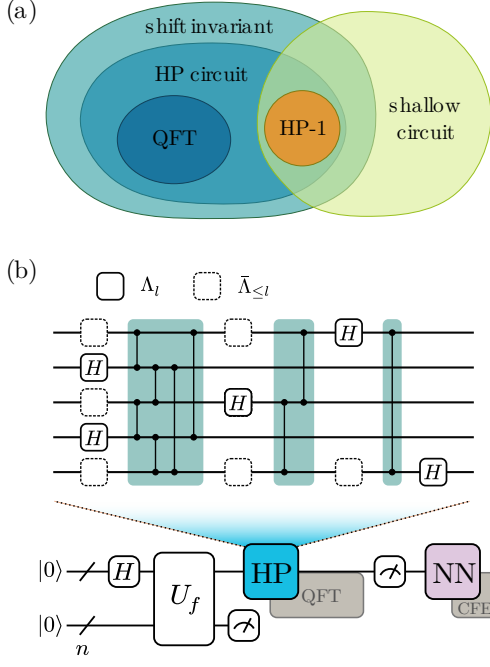


FIG. 1. **Shallow shift-invariant alternative to the QFT.** (a) Shift invariance is a critical property of the QFT. We design a circuit family, denoted as HP- L , that preserves the shift invariance. Within this broader family, HP-1 specifically serves as a shallow-depth circuit that is experimentally friendly. (b) In the proposed HSP algorithmic routine, the deep QFT block (the left gray gate) is directly replaced by the HP circuit. A neural network then processes the measurement outcomes, bypassing the standard continued fraction expansion (CFE) typically used in the standard HSP. The HP circuit is composed of Hadamard and phase gates, with Λ_l referring to the qubits with Hadamards applied in layer l .

(additional details about the HSP algorithm are provided in Appendix A).

We now define shift invariance [20] for HSP sampling and derive a necessary condition that any shift-invariant unitary must satisfy. For a subgroup $V \leq G$, in particular $V = \{|qr\rangle\}$, we denote the uniform superposition over its elements as $|V\rangle := \sum_{q \in \{0,1,2,\dots\}} |qr\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} |v\rangle$, for notational simplicity.

Definition 1 (Shift invariance). *A unitary U acting on the group register is shift-invariant for HSP sampling if*

$$|\langle x|U|V\rangle|^2 = |\langle x|U|c+V\rangle|^2, \quad \forall x, c \in G, \quad \forall V \leq G. \quad (1)$$

To identify shift-invariant circuits beyond the QFT, we first establish a necessary condition for shift invariance.

Proposition 1 (Necessary condition of shift invariance). *Suppose U is a unitary on the group register that satisfies the shift-invariance condition above. Then, for every*

$x, y \in G$ there exist phases $\theta_{xy} \in \mathbb{R}$ such that

$$\langle x|U|y\rangle = \frac{1}{\sqrt{|G|}} e^{i\theta_{xy}}. \quad (2)$$

Proof sketch of Proposition 1. The proof hinges on analyzing the shift-invariance condition for the trivial subgroup $V = \{0\}$. For $V = \{0\}$, the state $|V\rangle$ is the computational basis state $|0\rangle$, and the shifted state is $|c+V\rangle = |c\rangle$. The requirement (Eq.(1)) that $|\langle x|U|0\rangle|^2 = |\langle x|U|c\rangle|^2$ for all $x, c \in G$ implies that, for any fixed output x , the magnitude of every matrix element in the x -th row of U must be identical. The special case induces the uniform magnitude $\langle x|U|y\rangle = 1/\sqrt{|G|} e^{i\theta_{xy}}$. The full details of this argument are provided in Appendix B. $\square$

L -phase-block Hadamard-Phase circuits (HP): A shift-invariant circuit family.—Proposition 1 suggests a simple design principle for constructing shift-invariant circuits. The condition $|\langle x|U|y\rangle| = 1/\sqrt{|G|}$ requires flat amplitude magnitudes, which are naturally generated by Hadamard gates. The remaining freedom lies in the relative phases θ_{xy} , which can be adjusted by controlled-phase gates $\text{CP}(\theta) = |00\rangle\langle 00| + |01\rangle\langle 01| + |10\rangle\langle 10| + e^{i\theta}|11\rangle\langle 11|$ without changing these magnitudes. This motivates the Hadamard-phase (HP) circuit family we shall now introduce. As illustrated in Fig. 1 (b), the HP- L circuit is structured as an alternating sequence of Hadamard gates and phase blocks. To respect shift invariance, we enforce three rules for HP- L : (i) each qubit is acted on by a Hadamard gate exactly once; (ii) each qubit pair is connected by at most one controlled-phase gate; and (iii) each controlled-phase gate must come together with 2 Hadamards, one before and one after the phase gate. Here, “once” refers to the entire L -layer circuit rather than to a single layer.

To express the allowed circuits mathematically, we define two types of sets of qubits. Let H_i denote the Hadamard gate applied to qubit $i \in \{1, 2, \dots, n\}$. For each layer l , let Λ_l denote the set of qubits on which Hadamard gates are applied in that layer. The qubits that are still allowed in terms of having a Hadamard on them after layer l , i.e. those that have not had a Hadamard on them in or before layer l , are denoted as the set $\bar{\Lambda}_{\leq l}$.

In-between the Hadamard layers, the two sets are coupled via a controlled-phase block, denoted as $\text{CP}_{\Lambda_l, \bar{\Lambda}_{\leq l}}(\theta_{\Lambda_l, \bar{\Lambda}_{\leq l}})$, which applies pairwise operations $\text{CP}_{jk}(\theta_{jk})$ between each qubit $j \in \Lambda_l$ and $k \in \bar{\Lambda}_{\leq l}$. Building upon this notation, we formally define the circuit architecture:

Definition 2 (HP- L circuit). *A quantum circuit is classified as an HP- L circuit if it is constructed using the*

following layer-wise architecture

$$U_{\text{HP}}^{(L)}(\theta) = \bigotimes_{i \in \Lambda_L} H_i \prod_{l=L-1}^1 \left(\text{CP}_{\Lambda_l, \bar{\Lambda}_{\leq l}}(\theta_{\Lambda_l \bar{\Lambda}_{\leq l}}) \bigotimes_{i \in \Lambda_l} H_i \right), \quad (3)$$

where the controlled-phase block is given by

$$\text{CP}_{\Lambda_l, \bar{\Lambda}_{\leq l}}(\theta_{\Lambda_l \bar{\Lambda}_{\leq l}}) = \prod_{j \in \bar{\Lambda}_{\leq l}} \prod_{k \in \Lambda_l} \text{CP}_{jk}(\theta_{jk}). \quad (4)$$

The parameter set $\theta_{\Lambda_l \bar{\Lambda}_{\leq l}}$ is the collection of θ_{jk} , $j \in \Lambda_l$, $k \in \bar{\Lambda}_{\leq l}$.

In the following, we show that HP circuits preserve shift invariance. This implies that such invariance is not unique to the QFT, but is a feature that arises in a wider family of quantum circuits.

Theorem 1. *HP-L circuits are shift-invariant for $\mathbb{Z}_{2^n}$.*

The proof involves giving an analytical expression for the state evolved under an HP-L circuit (Eq.(3)) and checking that shift invariance is indeed respected (Appendix C).

HP-L for HSP over $\mathbb{Z}$: information on generator is retained.— For HSP over $\mathbb{Z}_{2^n}$, HP-0, namely a parallel Hadamard layer, gives an analytically solvable recovery procedure for the hidden subgroup generator 2^p . The key observation is that the first $n - p$ measurement bits are deterministically zero after applying $H^{\otimes n}$ to a coset state, allowing p to be read off from the zero pattern; the full calculation and sampling algorithm are given in Appendix D. However, the protocol HP-0 fails for general cyclic groups $\mathbb{Z}_q$ because the measurement distributions corresponding to different subgroups V are not distinguishable. That absence of distinguishability demonstrates that shift invariance alone is insufficient for successful hidden subgroup recovery, and that distinguishability should be imposed as an additional condition.

We use the discrete Fisher information (DFI) as a metric for distinguishability.

$$\text{DFI}(r, n) = \sum_{x=0}^{2^n-1} \frac{(\Pr(x | r+1) - \Pr(x | r))^2}{\Pr(x | r)}, \quad (5)$$

where x represents measurement outcomes, r is the hidden subgroup generator, and $\Pr(x | r)$ is the probability of measuring x when the subgroup generator is r .

For the standard QFT (Appendix E), we can calculate the DFI as

$$\text{DFI}_{\text{QFT}}(r, n) \approx \frac{4\pi^2}{9} \left(\frac{2^{2n}}{r^2} - 1 \right) \sim \mathcal{O}(2^{2n}). \quad (6)$$

Eq. (6) indicates that the generator r becomes easier to recover as the qubit number increases, which motivates us to limit r to $1 \ll r \ll 2^n$.

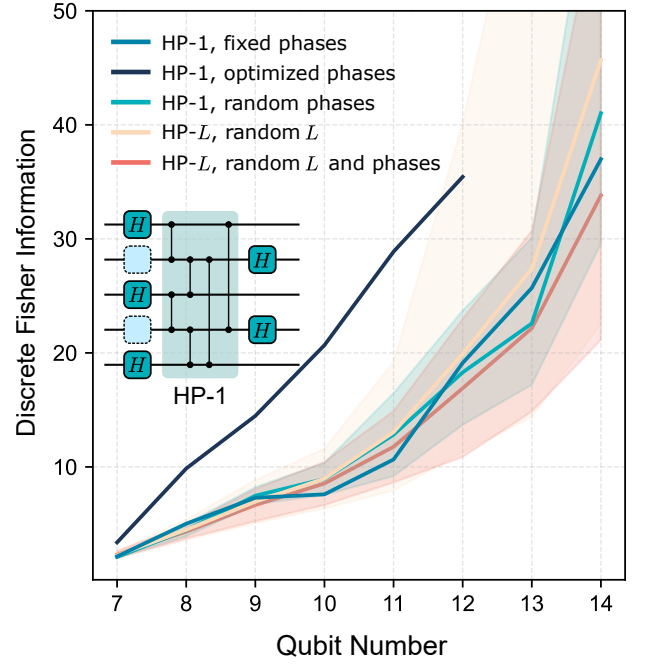


FIG. 2. **Average discrete Fisher information (DFI) increasing with the number of qubits.** Results are shown for several HP-family circuit settings. L denotes the number of phase blocks layers with HP-1 having a single phase block layer. Solid curves show the mean, with shaded bands indicating one standard deviation associated with random phase choices.

For HP circuits, the DFI is evaluated numerically. Specifically, for a fixed r , the circuit maps the periodic input components $\{|qr\rangle\}_q$ to a computational-basis distribution over outcomes $|b\rangle$, with the exact probabilities given by $\Pr(|b\rangle) = \left| \sum_q \langle b | U | qr \rangle \right|^2$. To evaluate the worst-case parameter sensitivity as the system scales, we consider the worst-case DFI: $\text{DFI}_{\min}(n) := \min_{2 \leq r \leq r_{\max}(n)} \text{DFI}(r, n)$, with $r_{\max}(n)^2 \ll 2^n$ in the asymptotic statement. Fig. 2 illustrates the minimum DFI against the qubit number n . (For the finite-size numerical scan, we use the square-root boundary proxy $r_{\max}(n) = \lfloor 2^{n/2} \rfloor$, matching the period scale used in the recovery experiments.)

Fig. 2 also compares the scaling of different HP circuit configurations, where solid curves and shaded bands represent the mean and standard deviation of $\text{DFI}_{\min}(n)$. The fixed-phase HP-1 circuit utilizes the skeleton, $\Lambda_1 = \{1, 3, 5, \dots\}$ and $\Lambda_2 = \{2, 4, 6, \dots\}$, with controlled-phase gates with phase $\frac{2\pi}{2^{|i-j|}}$ between qubits i and j . The optimized HP-1 circuit adopts the same skeleton but treats the gate phases as variational parameters, trained to maximize the $\text{DFI}_{\min}$. The deeper HP-L circuits are constructed by generating valid random skeletons Λ .

The proposed HP circuit family preserves an exponential growth trend of DFI over this square-root period window. Appendix F reports the corresponding ln-linear re-

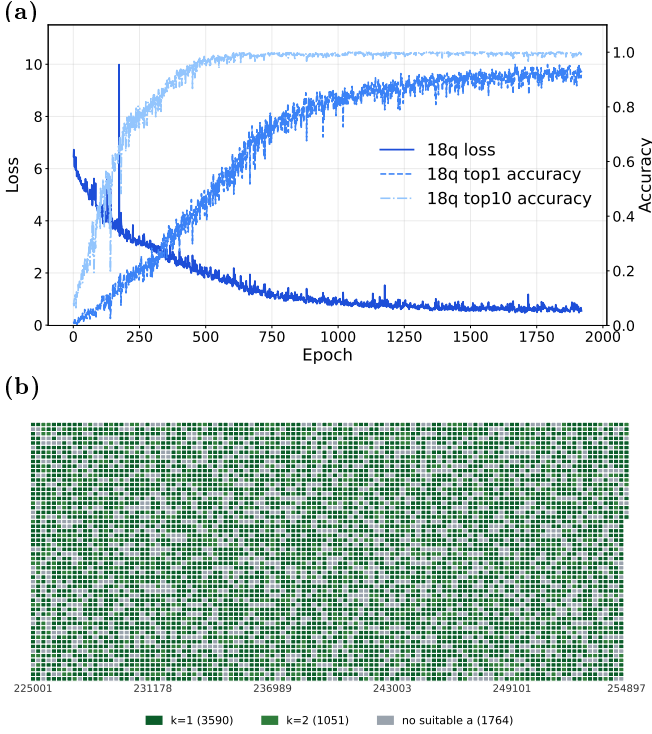


FIG. 3. **Numerical simulation shows successful Shor factoring via HP-1 and neural net post-processing.** (a) Validation loss, top-1 accuracy, and top-10 accuracy for neural-network period recovery on an 18-qubit instance. (b) Shor-type factorization on 18-qubit semiprime instances with $N = 225001, \dots, 254999$. Colour indicates the minimum number k of HP-1-based guesses needed for successful factorization, and grey entries indicate cases where no valid base a yields a detectable period within the chosen bound.

gression coefficients.

Efficient neural network for classical post-processing.—To avoid the exponential costs associated with processing full probability distributions, we employ a poly(n) Deep Sets neural network [19, 21] to predict the hidden subgroup generator directly from the raw measurement outcomes $\{b_i\}$. The input data is generated using the optimized HP-1 circuit evaluated in Fig. 2.

The outcomes are fed into a shared feature map ϕ , which maps each bitstring b_i to a feature vector within 2 linear layers and $16n$ feature dimension. The feature vectors are then summed to form an aggregated, permutation-invariant representation $\sum_i \phi(b_i)$. Subsequently, a beam-search decoder [22, 23] performs a top- k search to rank candidate periods. We use polynomial samples $m = 1024n^2$ to train the neural net. This architecture avoids exponential sample-space computations and outputs a ranked list of candidate generators for classical verification.

Replacing $\mathcal{O}(n^2)$ QFT with $\mathcal{O}(n)$ HP-1 in Shor's Algorithm.—We evaluate the optimized HP-1 circuit

in classical numerical simulations of Shor's algorithm for factoring with neural net classical post-processing. Shor's algorithm seeks the generator r satisfying $a^r \equiv 1 \pmod{N}$. We select the base a such that $1 \leq r \leq 2^{n/2}$, ensuring resolvability by the n -qubit register. We consider factoring 18-qubit semiprime instances with $N = 225001, \dots, 254999$.

The neural net trained successfully. As shown in Fig. 3(a), the neural network achieves rapid convergence when trained on an 18-qubit instance. Both the top-1 (correct period ranked first) and top-10 accuracies increase swiftly within a few epochs.

The combined HP-1 and neural net protocol recovers the subgroup generator in the tested instances. As illustrated in Fig. 3(b), each composite integer is colored by the minimum number of guesses k . Most solved instances are solved on the first attempt ($k = 1$), and the remaining solved instances require $k = 2$.

Noise simulation under a post-circuit global depolarizing channel demonstrates that our protocol with the optimized HP-1 circuit is robust against moderate environmental noise. The recovery accuracy remains stable up to noise-strength $\eta \approx 0.215$ for 10 qubits ($1 - \eta$ is the probability of applying the identity to the input state). Appendix G provides detailed settings, complete numerical results, and corresponding interpretations.

Summary and outlook.—We propose a hardware-efficient framework for solving the HSP by shallow, shift-invariant HP-1 circuits with a polynomial-scaling neural network. To verify the scalability of this protocol, we analyzed the growth model of the DFI. Finally, we evaluated our protocol on Shor's algorithm. The numerical simulations demonstrate that our approach successfully finds the hidden subgroup with high accuracy and remains robust against moderate depolarizing noise.

Realizing a practical HSP protocol further requires optimized implementations of the oracle U_f . Related strategies for reducing oracle and data-loading costs include Hamiltonian-learning-based state preparation that transfers the cost to classical preprocessing and uses $\mathcal{O}(1)$ oracle queries [24], space-time tradeoffs for non-Clifford gate reductions [25], and automatic oracle synthesis frameworks [26].

Apart from the $\mathcal{O}(n)$ scaling being hardware friendly, the reduction to grouped phase gates in HP circuits makes the phase operations easier to implement than individual phase gate execution. This hardware-level efficiency arises because consecutive phase operations can be merged into single continuous pulses across multiple experimental platforms, including in linear optics [27, 28], neutral-atom systems [29–31], trapped ions [32, 33], and NMR [34].

Acknowledgements.—We acknowledge support from The Guangdong Provincial Quantum Science Strategic

Initiative (Grant No. GDZX2503001) and the City University of Hong Kong (Proj. 9610623).

Code Availability.—All code relevant to this work is publicly available in the GitHub repository: <https://github.com/Fragcity/altqft>.

* These two authors contributed equally.

- [1] D. R. Simon, On the power of quantum computation, *SIAM Journal on Computing* **26**, 1474 (1997).
- [2] P. W. Shor, Algorithms for quantum computation: Discrete logarithms and factoring, in *Proceedings 35th Annual Symposium on Foundations of Computer Science* (IEEE, 1994) pp. 124–134.
- [3] P. W. Shor, Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer, *SIAM Journal on Computing* **26**, 1484 (1997).
- [4] A. Y. Kitaev, Quantum measurements and the abelian stabilizer problem (1995), arXiv:quant-ph/9511026 [quant-ph].
- [5] R. Jozsa, Quantum factoring, discrete logarithms, and the hidden subgroup problem, *Computing in Science & Engineering* **3**, 34 (2001).
- [6] C. Lomont, The hidden subgroup problem: Review and open problems (2004), arXiv:quant-ph/0411037 [quant-ph].
- [7] F. Liu, K. Bian, F. Meng, W. Zhang, and O. Dahlsten, Information compression via hidden subgroup quantum autoencoders, *npj Quantum Information* **10**, 74 (2024).
- [8] Y. Kim, A. Eddins, S. Anand, K. X. Wei, E. Van Den Berg, S. Rosenblatt, H. Nayfeh, Y. Wu, M. Zaletel, K. Temme, *et al.*, Evidence for the utility of quantum computing before fault tolerance, *Nature* **618**, 500 (2023).
- [9] Google Quantum AI and Collaborators, Quantum error correction below the surface code threshold, *Nature* **638**, 920 (2025).
- [10] D. Gao, D. Fan, C. Zha, *et al.*, Establishing a new benchmark in quantum computational advantage with 105-qubit Zhongzhong 3.0 processor, *Phys. Rev. Lett.* **134**, 090601 (2025).
- [11] R. Takagi, S. Endo, S. Minagawa, and M. Gu, Fundamental limits of quantum error mitigation, *npj Quantum Information* **8**, 114 (2022).
- [12] R. Takagi, H. Tajima, and M. Gu, Universal sampling lower bounds for quantum error mitigation, *Physical Review Letters* **131**, 210602 (2023).
- [13] P. Aumann, M. Fellner, D. Alber, M. Cykiert, C. Fleckenstein, R. ter Hoeven, L. Stenzel, R. J. Valencia-Tortora, and W. Lechner, Demonstrating record fidelity for the quantum fourier transform, arXiv preprint arXiv:2604.12465 (2026).
- [14] D. Coppersmith, An approximate fourier transform useful in quantum factoring (2002), arXiv:quant-ph/0201067 [quant-ph].
- [15] B. Park and D. Ahn, Reducing cnot count in quantum fourier transform for the linear nearest-neighbor architecture, *Scientific Reports* **13**, 8638 (2023).
- [16] E. Bäumer *et al.*, Quantum fourier transform using dynamic circuits, arXiv preprint arXiv:2403.09514 (2024).
- [17] W. Tang *et al.*, CutQC: using small quantum computers for large quantum circuit evaluations, in *Proceedings of the 26th ACM International Conference on Architectural Support for Programming Languages and Operating Systems* (2021) pp. 473–486.
- [18] T. Peng, A. W. Harrow, M. Ozols, and X. Wu, Simulating large quantum circuits on a small quantum computer, *Phys. Rev. Lett.* **125**, 150504 (2020).
- [19] M. Zaheer, S. Kottur, S. Ravanbakhsh, B. Póczos, R. R. Salakhutdinov, and A. J. Smola, Deep sets, *Advances in neural information processing systems* **30** (2017).
- [20] M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information* (Cambridge University Press, 2000).
- [21] H. Edwards and A. Storkey, Towards a neural statistician, arXiv preprint arXiv:1606.02185 (2016).
- [22] I. Sutskever, O. Vinyals, and Q. V. Le, Sequence to sequence learning with neural networks, in *Advances in neural information processing systems*, Vol. 27 (2014).
- [23] S. Wiseman and A. M. Rush, Sequence-to-sequence learning as beam-search optimization, in *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing* (2016) pp. 1296–1306.
- [24] M. Ramezani, S. A. Zargar, S. Salami, A. Bahrampour, and A. Bahrampour, Learning Hamiltonians for $\mathcal{O}(1)$ oracle-query quantum state preparation, arXiv preprint arXiv:2512.19181 (2025).
- [25] M. Yu, A. T. Calvino, M. Soeken, and G. De Micheli, Back-end-aware fault-tolerant quantum oracle synthesis, in *Proceedings of the 30th Asia and South Pacific Design Automation Conference (ASP-DAC)* (IEEE, 2025) pp. 930–937.
- [26] I.-T. Chen and C. Gupta, Towards efficient automatic oracle synthesis and resource estimation using QDK and QIR, in *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)* (IEEE, 2023).
- [27] J. Carolan, C. Harrold, C. Sparrow, E. Martín-López, N. J. Russell, J. W. Silverstone, P. J. Shadbolt, N. Matsuda, M. Oguma, M. Itoh, *et al.*, Universal linear optics, *Science* **349**, 711 (2015).
- [28] P. Kok, W. J. Munro, K. Nemoto, T. C. Ralph, J. P. Dowling, and G. J. Milburn, Linear optical quantum computing with photonic qubits, *Rev. Mod. Phys.* **79**, 135 (2007).
- [29] M. Saffman, T. G. Walker, and K. Mølmer, Quantum information with Rydberg atoms, *Reviews of modern physics* **82**, 2313 (2010).
- [30] S. J. Evered, D. Bluvstein, M. Kalinowski, S. Ebadi, T. Manovitz, H. Zhou, S. H. Li, A. A. Geim, T. T. Wang, N. Maskara, *et al.*, High-fidelity parallel entangling gates on a neutral-atom quantum computer, *Nature* **622**, 268 (2023).
- [31] M. Mohan, J. De Hond, and S. Kokkelmans, Parametrized multiqubit gates for neutral-atom quantum platforms, *Physical Review Applied* **23**, 054074 (2025).
- [32] D. Leibfried, B. DeMarco, V. Meyer, D. Lucas, M. Barrett, J. Britton, W. M. Itano, B. Jelenković, C. Langer, T. Rosenband, and D. J. Wineland, Experimental demonstration of a robust, high-fidelity geometric two ion-qubit phase gate, *Nature* **422**, 412 (2003).
- [33] T. R. Tan, J. P. Gaebler, R. Bowler, Y. Lin, J. D. Jost, D. Leibfried, and D. J. Wineland, Demonstration of a dressed-state phase gate for trapped ions, *Phys. Rev. Lett.* **110**, 263002 (2013).

- [34] L. M. Vandersypen and I. L. Chuang, NMR techniques for quantum control and computation, *Reviews of Modern Physics* **76**, 1037 (2004).
 [35] M. Ettinger, P. Høyer, and E. Knill, The quantum query

complexity of the hidden subgroup problem is polynomial, *Information Processing Letters* **91**, 43 (2004).

APPENDIX

A. Hidden Subgroup Problem and Fourier Sampling

For completeness, this section reviews the standard quantum algorithm for solving the Hidden Subgroup Problem (HSP) over finite abelian groups [4–6]. Formally, the HSP is defined as follows:

Definition 3 (Hidden Subgroup Problem). *Given a function $f : G \rightarrow X$ defined on a group G , with the promise that there exists a hidden subgroup $V \leq G$ such that for all $g_1, g_2 \in G$*

$$f(g_1) = f(g_2) \iff g_1 - g_2 \in V \quad (\text{A1})$$

The objective of this task is to identify the hidden subgroup V .

In the context of this paper, we restrict our attention to finite abelian groups. For such groups, the fundamental theorem of finite abelian groups allows us to express the group structure as a direct product of cyclic groups:

$$G \cong \mathbb{Z}_{p_1} \times \mathbb{Z}_{p_2} \times \cdots \times \mathbb{Z}_{p_k}. \quad (\text{A2})$$

The standard quantum procedure for solving the hidden subgroup problem begins by preparing a uniform superposition over all elements of the group G [4, 20]. We initialize the system in the state

$$|\psi\rangle = \frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle |0\rangle, \quad (\text{A3})$$

where $|G|$ is the number of elements of the group G . The first register holds the group elements, and the second register is initialized to a standard basis state. Next, we evaluate the quantum oracle corresponding to the function f , which acts on the two registers and entangles the group elements with their corresponding function values

$$|\psi\rangle = \frac{1}{\sqrt{|G|}} \sum_{g \in G} |g\rangle |f(g)\rangle. \quad (\text{A4})$$

Grouping the superposition terms according to cosets is illuminating, as the function f is constant within each coset of the hidden subgroup V and distinct across different cosets.

Before proceeding, it is useful to formalize the state notation for a subset. For any subset $S \subseteq G$, we define its corresponding normalized quantum state $|S\rangle$ as the uniform superposition over all elements within the set:

$$|S\rangle = \frac{1}{\sqrt{|S|}} \sum_{s \in S} |s\rangle. \quad (\text{A5})$$

Using this definition, the normalized state for a specific coset $c + V$ is directly given by

$$|c + V\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} |c + v\rangle. \quad (\text{A6})$$

By applying this coset state definition, the post-oracle state can be elegantly rewritten as a superposition of these orthogonal coset states, each perfectly entangled with its unique function evaluation:

$$|\psi\rangle = \sqrt{\frac{|V|}{|G|}} \sum_{c \in C} |c + V\rangle |f(c)\rangle, \quad (\text{A7})$$

where C is the set of all possible coset representatives. For simplicity, we assume that the second register is measured immediately after the oracle query. Measuring the second register yields a specific outcome $f(c)$ with a uniform probability of $|V|/|G|$. Consequently, the first register collapses into the corresponding pure coset state:

$$|\psi_{\text{post}}\rangle = |c + V\rangle. \quad (\text{A8})$$

From this point forward, our analysis will focus exclusively on the quantum state of this first register.

Whether the second register is physically measured to collapse the first register into a pure coset state $|c + V\rangle$, or traced out to leave it in a mixed state, the resulting Fourier-basis statistics remain identical [20]. The quantum Fourier transform produces a measurement distribution that is independent of the coset representative. For simplicity, we assume the second register is measured immediately after the oracle query, allowing our subsequent analysis to focus exclusively on the pure coset state:

$$|\psi_{\text{post}}\rangle = |c + V\rangle. \quad (\text{A9})$$

A direct measurement in the computational basis is uninformative. Indeed, using the definition of the coset state,

$$\langle g|c + V\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} \langle g|c + v\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} \delta_{g, c+v}. \quad (\text{A10})$$

For fixed c , the map $v \mapsto c + v$ is a bijection from V to the coset $c + V$. Therefore the sum $\sum_{v \in V} \delta_{g, c+v}$ equals 1 when $g \in c + V$ and equals 0 when $g \notin c + V$. Hence

$$\Pr[g | c, V] = |\langle g|c + V\rangle|^2 = \begin{cases} 1/|V|, & g \in c + V, \\ 0, & g \notin c + V, \end{cases} \quad (\text{A11})$$

so the observed element is uniform on the unknown coset. Averaging over uniformly random cosets gives the marginal distribution

$$\Pr[g] = \frac{1}{|G|}, \quad (\text{A12})$$

which contains no direct information about the hidden subgroup V .

To extract meaningful information, we must apply the QFT over the group G before measuring [4, 5]. Let $\chi_y(g)$ denote the group characters. The QFT maps a computational basis state $|g\rangle$ as follows:

$$\text{QFT } |g\rangle = \frac{1}{\sqrt{|G|}} \sum_{y \in G} \chi_y(g) |y\rangle. \quad (\text{A13})$$

Applying this transformation to our pure coset state $|c + V\rangle$ yields:

$$\begin{aligned} \text{QFT } |c + V\rangle &= \frac{1}{\sqrt{|V|}} \sum_{h \in V} \text{QFT } |c + h\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{v \in V} \sum_{y \in G} \chi_y(c + v) |y\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{y \in G} \chi_y(c) \left(\sum_{v \in V} \chi_y(v) \right) |y\rangle. \end{aligned} \quad (\text{A14})$$

By the orthogonality of group characters, the sum $\sum_{v \in V} \chi_y(v)$ evaluates to $|V|$ if the character χ_y is trivial on V (i.e., y belongs to the orthogonal subgroup $V^\perp$), and evaluates to 0 otherwise. Thus, the state simplifies to a superposition over $V^\perp$:

$$\text{QFT } |c + V\rangle = \sqrt{\frac{|V|}{|G|}} \sum_{y \in V^\perp} \chi_y(c) |y\rangle. \quad (\text{A15})$$

Measuring this state in the computational basis gives the probability of observing an outcome $y \in V^\perp$:

$$\Pr[y] = \left| \sqrt{\frac{|V|}{|G|}} \chi_y(c) \right|^2 = \frac{|V|}{|G|} |\chi_y(c)|^2 = \frac{|V|}{|G|}. \quad (\text{A16})$$

Crucially, because $|\chi_y(c)|^2 = 1$ for all group characters, the resulting measurement distribution is uniformly distributed over $V^\perp$ and is strictly independent of the random coset representative c . This shift invariance is the

fundamental property that allows the extraction of subgroup information by effectively removing the unknown shift at the probability level.

The QFT over G is defined using the character group $\hat{G}$ by

$$F_G|g\rangle = \frac{1}{\sqrt{|G|}} \sum_{\chi \in \hat{G}} \chi(g) |\chi\rangle, \quad (\text{A17})$$

where $\{|\chi\rangle : \chi \in \hat{G}\}$ is the Fourier basis and each character $\chi : G \rightarrow \mathbb{C}^\times$ is a group homomorphism. Here $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$ denotes the multiplicative group of nonzero complex numbers, so with additive notation on G one has $\chi(g_1 + g_2) = \chi(g_1)\chi(g_2)$. For finite abelian G , every character has unit modulus. Applying F_G to a coset state gives

$$\begin{aligned} F_G|c + V\rangle &= \frac{1}{\sqrt{|V|}} \sum_{v \in V} F_G|c + v\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{\chi \in \hat{G}} \sum_{v \in V} \chi(c + v) |\chi\rangle \\ &= \frac{1}{\sqrt{|V||G|}} \sum_{\chi \in \hat{G}} \chi(c) \left(\sum_{v \in V} \chi(v) \right) |\chi\rangle. \end{aligned} \quad (\text{A18})$$

The character sum over V satisfies

$$\sum_{v \in V} \chi(v) = \begin{cases} |V|, & \chi \in V^\perp, \\ 0, & \chi \notin V^\perp, \end{cases} \quad V^\perp = \{\chi \in \hat{G} : \chi(v) = 1 \text{ for all } v \in V\}. \quad (\text{A19})$$

Hence

$$F_G|c + V\rangle = \sqrt{\frac{|V|}{|G|}} \sum_{\chi \in V^\perp} \chi(c) |\chi\rangle. \quad (\text{A20})$$

The Fourier-basis measurement distribution is therefore

$$\Pr(\chi \mid c, V) = \begin{cases} |V|/|G|, & \chi \in V^\perp, \\ 0, & \chi \notin V^\perp, \end{cases} \quad (\text{A21})$$

which is independent of the coset representative c . The same formula holds for the mixed state ρ_V , since it is a convex combination of pure coset states with identical Fourier statistics.

Repeating the procedure therefore produces samples from $V^\perp$. In the finite abelian setting, classical post-processing reconstructs $V^\perp$ from the sample span and then recovers the hidden subgroup using the duality identity [4, 6, 35]

$$(V^\perp)^\perp = V. \quad (\text{A22})$$

Fourier sampling thus converts hidden translational structure in the oracle into explicit linear constraints in the dual group.

B. Necessary condition of shift invariance

Here we make explicit the probability-level shift-invariance condition used in Fourier sampling. Let G be a finite abelian group, and let $\{|g\rangle : g \in G\}$ denote the computational basis. For $c \in G$, define the shift operator

$$T_c|g\rangle = |g + c\rangle. \quad (\text{B1})$$

For a subgroup $V \leq G$, the normalized subgroup state is

$$|c + V\rangle = \frac{1}{\sqrt{|V|}} \sum_{v \in V} |c + v\rangle. \quad (\text{B2})$$

For HSP sampling, shift invariance requires that the measurement distribution after applying U is strictly independent of the coset representative:

$$|\langle x|U|V\rangle|^2 = |\langle x|UT_c|V\rangle|^2, \quad \forall x, c \in G, \forall V \leq G. \quad (\text{B3})$$

Operationally, this means $U|V\rangle$ and $U|c+V\rangle$ may differ by a relative phase, but they must yield identical probabilities in the computational basis. Any viable QFT replacement must therefore satisfy this condition.

Proposition (Necessary condition of shift invariance). *Suppose U is a unitary on the group register that satisfies Eq. (B3). Then, for every $x, y \in G$,*

$$|\langle x|U|y\rangle| = \frac{1}{\sqrt{|G|}}. \quad (\text{B4})$$

Equivalently, there exist phases $\theta_{xy} \in \mathbb{R}$ such that

$$\langle x|U|y\rangle = \frac{1}{\sqrt{|G|}} e^{i\theta_{xy}}. \quad (\text{B5})$$

To determine how this restricts the matrix elements of U , we can examine the trivial subgroup $V = \{0\}$. In this case, the subgroup state is simply the basis state $|0\rangle$, meaning the shift-invariance condition directly constrains the columns of U . Since the output probabilities for $|0\rangle$ and any shift $|g\rangle$ must be equal, for any fixed outcome x , all entries in the x -th row of U must share the same modulus. Unitarity then forces this common modulus to be exactly $1/\sqrt{|G|}$. This leads directly to Proposition 1.

Proof of Proposition 1. Take the trivial subgroup $V = \{0\}$. Then $|V\rangle = |0\rangle$ and $T_c|V\rangle = |c\rangle$ for every $c \in G$. Equation (B3) therefore implies

$$|\langle x|U|0\rangle|^2 = |\langle x|U|c\rangle|^2, \quad \forall x, c \in G. \quad (\text{B6})$$

Using the matrix element convention $U_{xy} = \langle x|U|y\rangle$, this equality shows that $|U_{x0}|^2 = |U_{xc}|^2$ for all $c \in G$. In other words, for each fixed row x , all entries have the same squared modulus. Denote this common value by a_x . Since U is unitary, the sum of the squared moduli across any row must be normalized to 1:

$$1 = \sum_{y \in G} |U_{xy}|^2 = |G|a_x. \quad (\text{B7})$$

Hence, $a_x = 1/|G|$ for every $x \in G$. Taking the square root yields the uniform modulus for all matrix elements:

$$|U_{xy}| = \frac{1}{\sqrt{|G|}}. \quad (\text{B8})$$

Because this modulus is strictly positive, every entry admits a polar decomposition

$$U_{xy} = \frac{1}{\sqrt{|G|}} e^{i\theta_{xy}}, \quad \theta_{xy} \in \mathbb{R}. \quad (\text{B9})$$

This proves the claim. $\square$

C. HP- L circuits are shift invariant over $\mathbb{Z}_{2^n}$

This appendix aims to rigorously prove that the HP- L circuit family satisfies the shift-invariance property. Establishing this theorem is of significant practical importance: it fundamentally guarantees that the output statistics obtained from this shallow circuit structure can be reliably used for the classical post-processing reconstruction of the hidden subgroup.

Theorem 1 (restated). *The HP- L circuit is shift-invariant over $\mathbb{Z}_{2^n}$.*

The core strategy of the proof is as follows: First, we explicitly calculate the sequential effect of each layer of the HP- L circuit on the initial basis state, deriving the analytical expression of the final quantum state. Next, we obtain the corresponding probability amplitudes by calculating the matrix elements of this output state in the computational basis. Finally, we compute the squared modulus of these amplitudes over the subgroup V and its coset $c + V$ respectively, using algebraic simplification to factor out a pure phase factor related to the coset representative c . Because the modulus of this pure phase factor is strictly 1, it naturally cancels out when calculating the measurement probabilities, thereby rigorously proving mathematically that the output probability distribution is completely independent of random shifts.

The $\bar{\Lambda}_{\leq l}$ is given by

$$\bar{\Lambda}_{\leq l} = \{1, 2, \dots, n\} \setminus \bigcup_{i \leq l} \Lambda_i. \quad (\text{C1})$$

The $\{\Lambda_l\}$ is a partition of index set $\{1, 2, \dots, n\}$, which means

$$\bigcup \Lambda_l = \{1, 2, \dots, n\}, \quad \Lambda_i \cap \Lambda_j = \emptyset. \quad (\text{C2})$$

Proof. The first step is to calculate the effect of the first layer, explicitly keeping the normalization factor $\frac{1}{\sqrt{2}}$ from the Hadamard gates:

$$\begin{aligned} & \text{CP}_{\Lambda_1, \bar{\Lambda}_{\leq 1}}(\theta_{\Lambda_1 \bar{\Lambda}_{\leq 1}}) \bigotimes_{i \in \Lambda_1} H_i |b_1 b_2 \dots b_n\rangle \\ &= \text{CP}_{\Lambda_1, \bar{\Lambda}_{\leq 1}}(\theta_{\Lambda_1 \bar{\Lambda}_{\leq 1}}) \bigotimes_{i \in \Lambda_1} \frac{1}{\sqrt{2}} (|0\rangle + e^{\pi i b_i} |1\rangle) \bigotimes_{k \in \bar{\Lambda}_1} |b_k\rangle \end{aligned} \quad (\text{C3})$$

$$= \bigotimes_{i \in \Lambda_1} \frac{1}{\sqrt{2}} \left(|0\rangle + \exp \left[\pi i b_i + i \sum_{j \in \bar{\Lambda}_{\leq 1}} \theta_{ij} b_j \right] |1\rangle \right) \bigotimes_{k \in \bar{\Lambda}_1} |b_k\rangle. \quad (\text{C4})$$

Note that since the qubits in $\bar{\Lambda}_{\leq 1}$ have not yet been acted upon by Hadamard gates, they remain in the computational basis.

Since the subsets Λ_l form a partition of the qubits:

$$\bigcup_{l=1}^L \Lambda_l = \{1, 2, \dots, n\}, \quad \Lambda_i \cap \Lambda_j = \emptyset, \quad (\text{C5})$$

and the subsets $\bar{\Lambda}_l$ are defined as the remaining qubits that have not yet been acted on by Hadamard gates:

$$\bar{\Lambda}_{\leq l} = \{1, 2, \dots, n\} \setminus \left(\bigcup_{k=1}^l \Lambda_k \right), \quad (\text{C6})$$

the sequential action of all L layers ensures every qubit i is acted on by a Hadamard gate exactly once. Let $l(i)$ denote the specific stage where $i \in \Lambda_{l(i)}$. The final state after the full circuit becomes:

$$U_{\text{HP}}^{(L)}(\theta) |b_1 b_2 \dots b_n\rangle = \bigotimes_{i=1}^n \frac{1}{\sqrt{2}} \left(|0\rangle + \exp \left[\pi i b_i + i \sum_{j \in \bar{\Lambda}_{l(i)}} \theta_{ij} b_j \right] |1\rangle \right). \quad (\text{C7})$$

To find the matrix element, we take the inner product with $\langle b'_1 b'_2 \dots b'_n |$. For each qubit i , the $|1\rangle$ component is selected if and only if $b'_i = 1$, which mathematically corresponds to multiplying the entire phase exponent by b'_i . Thus, we obtain:

$$\begin{aligned} & \langle b'_1 b'_2 \dots b'_n | U_{\text{HP}}^{(L)}(\theta) |b_1 b_2 \dots b_n\rangle \\ &= \frac{1}{\sqrt{2}^n} \prod_{i=1}^n \exp \left[\pi i b_i b'_i + i \sum_{j \in \bar{\Lambda}_{l(i)}} \theta_{ij} b'_i b_j \right] \end{aligned} \quad (\text{C8})$$

$$= \frac{1}{\sqrt{2^n}} \exp \left[i\pi \sum_{i=1}^n b_i b'_i + i \sum_{l=1}^{L-1} \sum_{i \in \Lambda_l} \sum_{j \in \bar{\Lambda}_{\leq l}} \theta_{ij} b'_i b_j \right]. \quad (\text{C9})$$

To simplify the notation for the matrix elements while respecting the sequential asymmetry of the circuit (where each qubit pair interacts only once), we define a directed effective phase matrix $\tilde{\Theta}$ whose elements are given by:

$$\tilde{\theta}_{ij} = \begin{cases} \theta_{ij}, & \text{if } i \in \Lambda_{l(i)}, j \in \bar{\Lambda}_{l(i)} \text{ (i.e., } l(i) < l(j)) \\ \pi, & \text{if } i = j \\ 0, & \text{otherwise (i.e., } l(i) \geq l(j)). \end{cases} \quad (\text{C10})$$

Using this effective matrix, the exponent can be compactly written as a full summation over all i, j :

$$\langle b'_1 b'_2 \cdots b'_n | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | b_1 b_2 \cdots b_n \rangle = \frac{1}{\sqrt{2^n}} \exp \left[i \sum_{i,j} \tilde{\theta}_{ij} b'_i b_j \right]. \quad (\text{C11})$$

The shift invariance requires that the probability distribution over the subgroup V is identical to its cosets. First, we compute the squared probability amplitude over V :

$$\left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | V \rangle \right|^2 = \frac{1}{2^n |V|} \left| \sum_{y \in V} \exp \left[i \sum_{i,j} \tilde{\theta}_{ij} x_i y_j \right] \right|^2. \quad (\text{C12})$$

Since $V \leq \mathbb{Z}_{2^n}$, $V = \{q2^s \mid q \in \mathbb{Z}\}$ for certain $s \leq n$. Assuming a standard binary encoding where the s least significant bits correspond to the trailing qubits, $y \in V$ can be denoted as $y = y_1 y_2 \cdots y_{n-s} 0 \cdots 0$, with $y_i \in \{0, 1\}$. Similarly, the coset $c + V = \{q2^s + c \mid 0 \leq c < 2^s, q \in \mathbb{Z}\}$, and its elements $y \in c + V$ can be denoted by matching the trailing bits to the binary representation of c , yielding $y = y_1 y_2 \cdots y_{n-s} c_{n-s+1} \cdots c_n$, with $y_i, c_k \in \{0, 1\}$.

Therefore, we can expand the squared amplitude for the subgroup V :

$$\begin{aligned} & \left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | V \rangle \right|^2 \\ &= \frac{1}{2^n |V|} \left| \sum_{y \in V} \exp \left[i \sum_{i=1}^n \sum_{j=1}^n \tilde{\theta}_{ij} x_i y_j \right] \right|^2 \end{aligned} \quad (\text{C13})$$

$$= \frac{1}{2^n 2^{n-s}} \left| \sum_{y_1, \dots, y_{n-s} \in \{0,1\}} \exp \left[i \sum_{i=1}^n \sum_{j=1}^{n-s} \tilde{\theta}_{ij} x_i y_j \right] \right|^2. \quad (\text{C14})$$

Evaluating the same expression for the coset $c + V$, we factor out the terms involving the constant bits c_k :

$$\begin{aligned} & \left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | c + V \rangle \right|^2 \\ &= \frac{1}{2^n |c + V|} \left| \sum_{y \in c + V} \exp \left[i \sum_{i=1}^n \sum_{j=1}^n \tilde{\theta}_{ij} x_i y_j \right] \right|^2 \end{aligned} \quad (\text{C15})$$

$$= \frac{1}{2^n 2^{n-s}} \left| \exp \left[i \sum_{i=1}^n \sum_{k=n-s+1}^n \tilde{\theta}_{ik} x_i c_k \right] \sum_{y_1, \dots, y_{n-s} \in \{0,1\}} \exp \left[i \sum_{i=1}^n \sum_{j=1}^{n-s} \tilde{\theta}_{ij} x_i y_j \right] \right|^2 \quad (\text{C16})$$

$$= \frac{1}{2^n 2^{n-s}} \left| \exp \left[i \sum_{i=1}^n \sum_{k=n-s+1}^n \tilde{\theta}_{ik} x_i c_k \right] \right|^2 \left| \sum_{y_1, \dots, y_{n-s} \in \{0,1\}} \exp \left[i \sum_{i=1}^n \sum_{j=1}^{n-s} \tilde{\theta}_{ij} x_i y_j \right] \right|^2. \quad (\text{C17})$$

Because the modulus of the extracted pure phase factor strictly equals 1, it completely cancels out, yielding:

$$\left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | c + V \rangle \right|^2 = \frac{1}{2^n 2^{n-s}} \left| \sum_{y_1, \dots, y_{n-s} \in \{0,1\}} \exp \left[i \sum_{i=1}^n \sum_{j=1}^{n-s} \tilde{\theta}_{ij} x_i y_j \right] \right|^2 \quad (\text{C18})$$

$$= \left| \langle x | U_{\text{HP}}^{(L)}(\boldsymbol{\theta}) | V \rangle \right|^2. \quad (\text{C19})$$

This completes the proof. $\square$

D. Sampling calculation and recovery guarantee for the HP – 0 case

We now give the calculation underlying the HP-0 discussion in the main text. Consider $G \simeq \mathbb{Z}_{2^n}$ represented by n computational-basis qubits. Because $\mathbb{Z}_{2^n}$ is a cyclic group of order 2^n , any of its subgroups must be generated by a divisor of 2^n . Thus, the hidden subgroup is necessarily of the form $V = \langle 2^p \rangle$ for some integer $0 \leq p \leq n$. Since every element $h \in V$ is a multiple of 2^p , its n -bit binary representation naturally ends with p zeros. Under this standard convention, an element h takes the form

$$h = h_1 h_2 \cdots h_{n-p} 0 \cdots 0. \quad (\text{D1})$$

The subgroup generated by 2^p is

$$V = \{q2^p : q = 0, \dots, 2^{n-p} - 1\}, \quad (\text{D2})$$

so $|V| = 2^{n-p}$. Since adding an element of V can vary the first $n-p$ bits, each coset has a representative of the form

$$c = 0 \cdots 0 c_{n-p+1} \cdots c_n. \quad (\text{D3})$$

Thus a normalized coset state can be written as

$$|c + V\rangle = \frac{1}{\sqrt{2^{n-p}}} \sum_{h_1, \dots, h_{n-p} \in \{0,1\}} |h_1 \cdots h_{n-p} c_{n-p+1} \cdots c_n\rangle. \quad (\text{D4})$$

For an n -bit string x , applying Hadamard gates to all n qubits gives

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} (-1)^{k \cdot x} |k\rangle, \quad (\text{D5})$$

where $k \cdot x = \sum_{i=1}^n k_i x_i$ denotes the standard bitwise inner product of the binary strings k and x . Applying this to Eq. (D4) gives

$$\begin{aligned} H^{\otimes n} |c + V\rangle &= \frac{1}{\sqrt{2^{n-p}}} \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} \sum_{h_1, \dots, h_{n-p} \in \{0,1\}} (-1)^{\sum_{i=1}^{n-p} k_i h_i + \sum_{i=n-p+1}^n k_i c_i} |k\rangle \\ &= \frac{1}{\sqrt{2^{n-p}}} \frac{1}{\sqrt{2^n}} \sum_{k \in \{0,1\}^n} (-1)^{\sum_{i=n-p+1}^n k_i c_i} \prod_{i=1}^{n-p} \left(\sum_{h_i \in \{0,1\}} (-1)^{k_i h_i} \right) |k\rangle. \end{aligned} \quad (\text{D6})$$

The inner product factor is zero unless $k_1 = \cdots = k_{n-p} = 0$, in which case it equals 2^{n-p} . Therefore

$$H^{\otimes n} |c + V\rangle = \frac{1}{\sqrt{2^p}} \sum_{k_{n-p+1}, \dots, k_n \in \{0,1\}} (-1)^{\sum_{i=n-p+1}^n k_i c_i} |0\rangle^{\otimes (n-p)} |k_{n-p+1} \cdots k_n\rangle. \quad (\text{D7})$$

The coset representative c is eliminated in Eq. (D7). Consequently, the measurement distribution is independent of c . The first $n-p$ bits are always zero, while the remaining p bits are uniformly random.

This immediately gives a simple reconstruction procedure for the idealized $\mathbb{Z}_{2^n}$ HSP. Each application of the oracle-and-measurement routine prepares a random coset state, and Eq. (D7) shows that subsequently applying $H^{\otimes n}$ converts that coset state into a sample whose first $n-p$ bits are deterministically zero while the remaining p bits are unbiased. The resulting sampling-and-recovery routine is summarized in Algorithm 1.

We note that the physical measurement of the second register in Algorithm 1 is not strictly necessary. By the principle of deferred measurement, tracing out the second register leaves the first register in a mixed state of all possible cosets. Since the measurement distribution is strictly independent of the coset representative c , this yields

Algorithm 1: HP – 0 Sampling for $V \leq \mathbb{Z}_{2^n}$

Input: Oracle access to an HSP instance over $\mathbb{Z}_{2^n}$, sample count m
Output: Hidden subgroup $V = \langle 2^p \rangle$

```

1 Initialize an empty list of samples  $\mathcal{B}$ ;
2 for  $i = 1$  to  $m$  do
3   Prepare the uniform superposition  $\frac{1}{\sqrt{2^n}} \sum_{g \in \mathbb{Z}_{2^n}} |g\rangle |0\rangle$ ;
4   Query the oracle (measuring the second register to obtain a coset state  $|c_i + V\rangle$  is optional);
5   Apply  $H^{\otimes n}$  to the first register;
6   Measure the first register and record the bit string  $b^{(i)} \in \{0, 1\}^n$ ;
7   Append  $b^{(i)}$  to  $\mathcal{B}$ ;
8 end
9 Find the minimum index  $j^*$  (where  $j = 1$  is the most significant bit) such that  $b_{j^*}^{(i)} = 1$  for some sample in  $\mathcal{B}$ ;
10 if such a  $j^*$  exists then
11   Let  $p = n - j^* + 1$ ;
12 else
13   Let  $p = 0$ ;
14 end
15 return  $V = \langle 2^p \rangle$ ;

```

identical measurement statistics. This device-friendly feature avoids the need for mid-circuit measurements, which is highly advantageous for implementations on near-term quantum hardware.

Algorithm 1 is correct because, under the bit-ordering convention above, the free positions occur exactly in the final p bits. Equivalently, the measurement data have the form

$$0 \cdots 0 * \cdots *, \quad (\text{D8})$$

where each star denotes an unbiased bit. Therefore, the classical post-processing only needs to identify the leftmost bit position that ever evaluates to 1 across all m samples. This single position j^* uniquely determines p , and hence determines the subgroup $V = \langle 2^p \rangle$.

The sample complexity only scales with $\ln(1/\epsilon)$. Under this leftmost-1 decoding strategy, the algorithm only fails if the most significant free bit (at position $n - p + 1$) is measured as 0 in all m independent samples. Because this specific bit is uniformly random, the probability of it being zero m times in a row is exactly 2^{-m} . Thus $m \geq \lceil \ln(1/\epsilon) / \ln 2 \rceil$ samples are sufficient for Algorithm 1 to perfectly identify the hidden subgroup V with probability at least $1 - \epsilon$. This tight bound elegantly removes the union-bound dependence on n , yielding an optimized sample complexity.

E. Fisher information for subgroup-generator estimation

To benchmark the proposed circuit family, we treat the hidden subgroup generator r as the parameter to be inferred from the measurement distribution produced by the standard QFT. Consider the truncated periodic state

$$R := \left\lfloor \frac{2^n - 1}{r} \right\rfloor + 1, \quad (\text{E1})$$

where R is the number of supported multiples of r , and

$$|\psi_r\rangle := \frac{1}{\sqrt{R}} \sum_{q=0}^{R-1} |qr\rangle. \quad (\text{E2})$$

Thus $q = 0, \dots, R-1$ labels precisely the integers qr contained in the 2^n -dimensional register. For the 2^n -point QFT,

$$|j\rangle \mapsto \frac{1}{\sqrt{2^n}} \sum_{k=0}^{2^n-1} e^{2\pi i j k / 2^n} |k\rangle, \quad (\text{E3})$$

and therefore

$$|\psi_r\rangle \mapsto \frac{1}{\sqrt{R2^n}} \sum_{k=0}^{2^n-1} \left(\sum_{q=0}^{R-1} e^{2\pi i q r k / 2^n} \right) |k\rangle. \quad (\text{E4})$$

The corresponding measurement probability is

$$\Pr(k|r) = \frac{1}{R2^n} \left| \sum_{q=0}^{R-1} e^{2\pi i q r k / 2^n} \right|^2 = \frac{1}{R2^n} \left(\frac{\sin(\pi R r k / 2^n)}{\sin(\pi r k / 2^n)} \right)^2. \quad (\text{E5})$$

In the continuum approximation, the Fisher-information calculation reduces to a Dirichlet-kernel moment estimate. The result needed for the main-text comparison is summarized in the following lemma.

Lemma 1 (QFT Fisher-information). *In the regime $1 \ll r \ll 2^n$, the QFT Fisher information is approximated by*

$$\text{DFI}_{\text{QFT}}(r, n) \approx \frac{4\pi^2}{9} \left(\frac{2^{2n}}{r^2} - 1 \right). \quad (\text{E6})$$

Proof. In the regime $r \ll 2^n$, introduce the continuum measurement coordinate

$$x := \frac{\pi k}{2^n} \in [0, \pi]. \quad (\text{E7})$$

The corresponding continuum approximation to the QFT probability density is

$$f(x, r) := \frac{1}{\pi R} \left(\frac{\sin(Rrx)}{\sin(rx)} \right)^2. \quad (\text{E8})$$

The QFT Fisher information with respect to the generator r is

$$\text{DFI}_{\text{QFT}}(r, n) = \int_0^\pi f(x, r) \left[\frac{\partial}{\partial r} \ln f(x, r) \right]^2 dx, \quad (\text{E9})$$

with score function

$$s_r(x) := \frac{\partial}{\partial r} \ln f(x, r) = 2Rx \cot(Rrx) - 2x \cot(rx), \quad s_r(x) = \text{score for the generator } r. \quad (\text{E10})$$

Define

$$D_R(u) := \frac{\sin(Ru)}{\sin u}, \quad D_R(u) = \text{order-}R \text{ Dirichlet kernel}. \quad (\text{E11})$$

Then $f(x, r) = D_R(rx)^2 / (\pi R)$, and the score function implies

$$f(x, r) \left[\frac{\partial}{\partial r} \ln f(x, r) \right]^2 = \frac{4x^2}{\pi R} [D'_R(rx)]^2. \quad (\text{E12})$$

Thus the Fisher-information integral is reduced to the square of a finite trigonometric polynomial. Once that polynomial is expanded into cosine modes, the remaining calculation involves only the moments of x^2 against those modes.

We begin by expressing the Dirichlet kernel $D_R(u)$ in its exponential sum form. By shifting the summation index to center the phases, it can be written as a symmetric finite series:

$$D_R(u) = \frac{\sin(Ru)}{\sin u} = \sum_{m=0}^{R-1} e^{i(R-1-2m)u}, \quad (\text{E13})$$

Differentiating this series term-by-term with respect to u yields:

$$D'_R(u) = i \sum_{m=0}^{R-1} (R-1-2m) e^{i(R-1-2m)u}. \quad (\text{E14})$$

To compute the squared magnitude $[D'_R(u)]^2$, we multiply this sum by itself. By collecting terms that produce the same relative frequency difference, we can perfectly recast the product as an even cosine series:

$$[D'_R(u)]^2 = A_0 + 2 \sum_{j=1}^{R-1} A_j \cos(2ju), \quad (\text{E15})$$

where A_j denotes the cosine-series coefficient at frequency $2j$. These coefficients are obtained by evaluating the convolution sum over amplitudes with frequency offset j . To evaluate this sum explicitly, we expand the summand as a quadratic in m and apply the standard summation formulas for $\sum m$ and $\sum m^2$:

$$\begin{aligned}
A_j &= - \sum_{m=0}^{R-1-j} (R-1-2m)(2j-R+1+2m) \\
&= \sum_{m=0}^{R-1-j} [4m^2 - 4(R-1-j)m + (R-1-j)^2 - j^2] \\
&= \frac{R-j}{3} [(R-j-1)^2 + 2(R-j-1) - 3j^2] \\
&= \frac{(R-j)(R^2 - 2Rj - 2j^2 - 1)}{3}, \quad j = 0, \dots, R-1.
\end{aligned} \tag{E16}$$

In particular, setting $j = 0$ isolates the non-oscillatory constant term:

$$A_0 = \frac{R(R^2 - 1)}{3}. \tag{E17}$$

Recall from our earlier derivation of the score function that the Fisher information integral takes the form:

$$\text{DFI}_{\text{QFT}}(r, n) = \frac{4}{\pi R} \int_0^\pi x^2 [D'_R(rx)]^2 dx, \tag{E18}$$

Substituting our exact cosine expansion of $[D'_R(rx)]^2$ into this integral distributes the integration across the individual frequency modes:

$$\text{DFI}_{\text{QFT}}(r, n) = \frac{4}{\pi R} \left[A_0 \int_0^\pi x^2 dx + 2 \sum_{j=1}^{R-1} A_j \int_0^\pi x^2 \cos(2jrx) dx \right]. \tag{E19}$$

The problem is now reduced to evaluating two elementary definite integrals over $[0, \pi]$. Define the non-oscillatory and oscillatory moments by

$$M_0 := \int_0^\pi x^2 dx = \frac{\pi^3}{3}, \tag{E20}$$

$$\begin{aligned}
M_j(r) &:= \int_0^\pi x^2 \cos(2jrx) dx \\
&= \frac{4\pi jr \cos(2\pi jr) + (4\pi^2 j^2 r^2 - 2) \sin(2\pi jr)}{(2jr)^3}, \quad j \geq 1.
\end{aligned} \tag{E21}$$

Here M_0 is the non-oscillatory second moment, and $M_j(r)$ is the oscillatory second moment at mode j . Inserting these evaluated moments, along with the explicit expressions for A_0 and A_j , back into the expanded Fisher information equation gives:

$$\begin{aligned}
\text{DFI}_{\text{QFT}}(r, n) &= \frac{4}{\pi R} \left[\frac{R(R^2 - 1)}{3} \cdot \frac{\pi^3}{3} + \frac{2}{3} \sum_{j=1}^{R-1} (R-j)(R^2 - 2Rj - 2j^2 - 1) M_j(r) \right] \\
&= \frac{4\pi^2}{9} (R^2 - 1) + \frac{8}{3\pi R} \sum_{j=1}^{R-1} (R-j)(R^2 - 2Rj - 2j^2 - 1) M_j(r),
\end{aligned} \tag{E22}$$

which is the stated exact formula.

Finally, we analyze the asymptotic behavior in the standard regime of interest, $1 \ll r \ll R$. In this limit, the hidden parameter is large enough to suppress oscillations, yet small enough that its multiples are heavily supported in the register. The explicit form of the oscillatory moments shows that $M_j(r) = \mathcal{O}(r^{-2})$ for any fixed mode j . Since the leading constant term scales as $\mathcal{O}(R^2)$ and the summation over the modes scales proportionally to $\mathcal{O}(R^2 r^{-2})$,

the relative magnitude of the oscillatory corrections is heavily suppressed by a factor of $1/r^2$. Because these high-frequency corrections are strictly subleading in this large- r regime, the behavior is overwhelmingly governed by the constant contribution. Keeping only this non-oscillatory term and substituting $R \approx 2^n/r$ therefore gives:

$$\text{DFI}_{\text{QFT}}(r, n) \approx \frac{4\pi^2}{9} \left(\frac{2^{2n}}{r^2} - 1 \right). \quad (\text{E23})$$

This proves the lemma. $\square$

Lemma 2 (HP-1 period-law overlap). *For the fixed-phase HP-1 circuit, let Pr_s and Pr_t denote the coherent output probability laws associated with two candidate periods s, t . Their overlap is*

$$\sum_x \text{Pr}_s(x) \text{Pr}_t(x) = \quad (\text{E24})$$

Proof. Set $N := 2^n$. For the fixed-phase HP-1 circuit, let $C = C_{\text{HP1}} \in \mathbb{R}^{n \times n}$ denote the HP-1 phase matrix normalized by

$$\langle x | U_{\text{HP}}^{(1)} | y \rangle = \frac{1}{\sqrt{N}} \exp(\pi i x^T C y), \quad x, y \in \{0, 1\}^n. \quad (\text{E25})$$

Equivalently, C is the directed fixed-phase HP-1 matrix with nonzero entries

$$C_{ij} = \begin{cases} \frac{1}{2^{|i-j|}}, & i = j \text{ or } (i \in \Lambda_1, j \in \Lambda_2), \\ 0, & \text{otherwise,} \end{cases} \quad (\text{E26})$$

For a candidate period s , define

$$R_s := |\{q \in \mathbb{Z}_{\geq 0} : 0 \leq qs < N\}| = \left\lfloor \frac{N-1}{s} \right\rfloor + 1. \quad (\text{E27})$$

Let $\mathbf{b}_m \in \{0, 1\}^n$ denote the n -bit binary representation of the integer m . Using Eq. (E25), the probability law Pr_s is

$$\begin{aligned} \text{Pr}_s(x) &= \left| \frac{1}{\sqrt{R_s}} \sum_{q=0}^{R_s-1} \langle x | U_{\text{HP}}^{(1)} | \mathbf{b}_{qs} \rangle \right|^2 \\ &= \frac{1}{NR_s} \left| \sum_{q=0}^{R_s-1} \exp(\pi i x^T C \mathbf{b}_{qs}) \right|^2, \quad x \in \{0, 1\}^n. \end{aligned} \quad (\text{E28})$$

Substituting Eq. (E28) and the corresponding expression for Pr_t into Eq. (E24) gives

$$\begin{aligned} \sum_x \text{Pr}_s(x) \text{Pr}_t(x) &= \frac{1}{N^2 R_s R_t} \sum_{x \in \{0, 1\}^n} \left| \sum_{q=0}^{R_s-1} \exp(\pi i x^T C \mathbf{b}_{qs}) \right|^2 \\ &\quad \times \left| \sum_{\ell=0}^{R_t-1} \exp(\pi i x^T C \mathbf{b}_{\ell t}) \right|^2. \end{aligned} \quad (\text{E29})$$

For $v \in \mathbb{Z}^n$, the associated conjugate-pair identity defines the real cosine kernel $\Omega(v)$ by

$$\prod_{a=1}^n [1 + e^{\pi i (Cv)_a}] + \prod_{a=1}^n [1 + e^{-\pi i (Cv)_a}] = 2N \cos \left[\frac{\pi}{2} \sum_{a=1}^n (Cv)_a \right] \prod_{a=1}^n \cos \left[\frac{\pi}{2} (Cv)_a \right] =: 2N \Omega(v). \quad (\text{E30})$$

For fixed q, q', ℓ, ℓ' , set

$$\Delta_{q, q', \ell, \ell'} := \mathbf{b}_{qs} - \mathbf{b}_{q's} + \mathbf{b}_{\ell t} - \mathbf{b}_{\ell' t}.$$

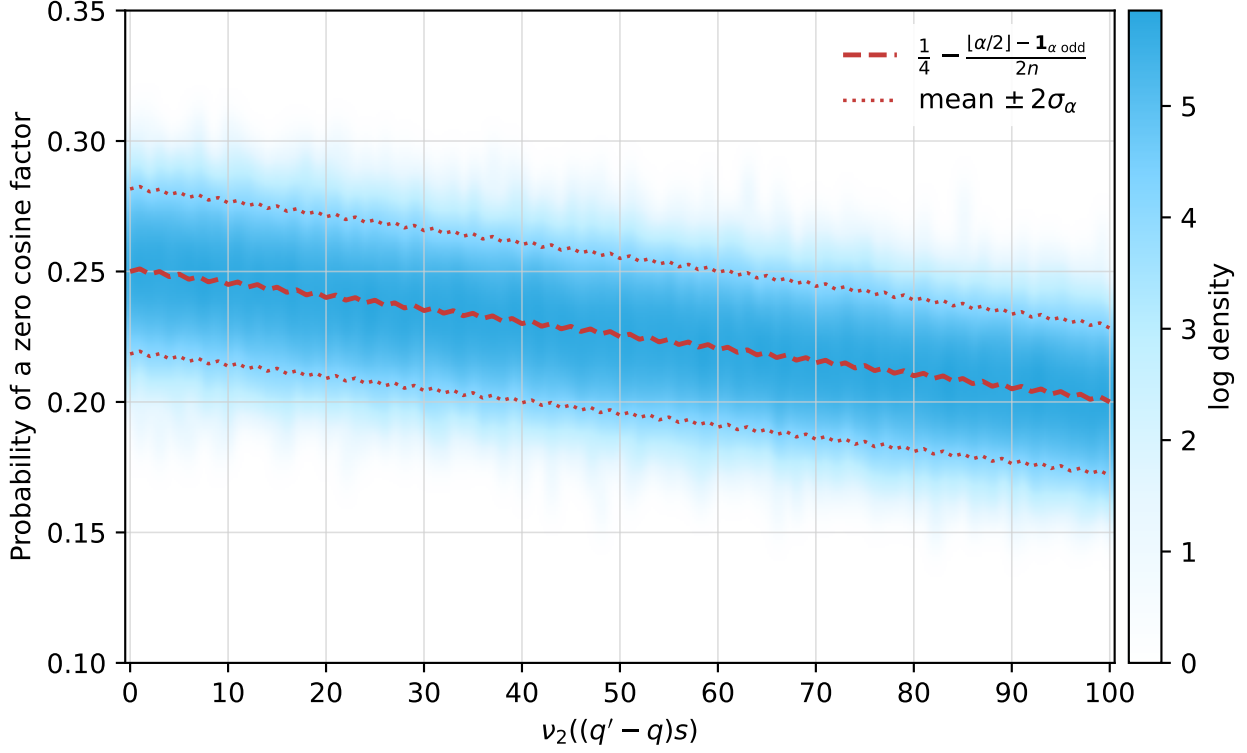


FIG. 4. Gaussian-splat density plot of the empirical probability that $\cos[\pi(C\Delta_{q,q',0,0})_a/2] = 0$, grouped by $\alpha = \nu_2((q' - q)s)$. The data use $n = 500$ and 500 stratified samples for each $\alpha = 0, \dots, 100$. The dashed curve is the empirical mean model, and the dotted curves show its $\pm 2\sigma_\alpha$ band.

Expanding the two squared moduli in Eq. (E29), exchanging the order of summation, and substituting the x -sum kernel explicitly gives

$$\begin{aligned}
 \sum_x \Pr_s(x) \Pr_t(x) &= \frac{1}{N^2 R_s R_t} \sum_{q,q'=0}^{R_s-1} \sum_{\ell,\ell'=0}^{R_t-1} \prod_{a=1}^n [1 + \exp(\pi i (C\Delta_{q,q',\ell,\ell'})_a)] \\
 &= \left[\frac{1}{N} + \frac{2}{NR_s} \sum_{0 \leq q < q' < R_s} \Omega(\Delta_{q,q',0,0}) + \frac{2}{NR_t} \sum_{0 \leq \ell < \ell' < R_t} \Omega(\Delta_{0,0,\ell,\ell'}) \right. \\
 &\quad \left. + \frac{2}{NR_s R_t} \sum_{0 \leq q < q' < R_s} \sum_{0 \leq \ell < \ell' < R_t} [\Omega(\Delta_{q,q',\ell,\ell'}) + \Omega(\Delta_{q,q',\ell',\ell})] \right]. \quad (\text{E31})
 \end{aligned}$$

Here $\Delta_{q,q',0,0} = \mathbf{b}_{qs} - \mathbf{b}_{q's}$ and $\Delta_{0,0,\ell,\ell'} = \mathbf{b}_{\ell t} - \mathbf{b}_{\ell' t}$, since the paired diagonal difference is zero. The first equality is the fully substituted form of the x -sum: no additional kernel notation is being used. The passage from the first line of Eq. (E31) to the second line comes from grouping ordered indices into the smallest real orbits. The conjugate-pair identity in Eq. (E30) is applied to each conjugate pair. If $q = q'$ and $\ell = \ell'$, then $\Delta = 0$, the product equals N , and the $R_s R_t$ such terms give the first normalized contribution $1/N$ inside the brackets. If $q \neq q'$ while $\ell = \ell'$, then for each unordered pair $q < q'$ and each of the R_t diagonal values of ℓ , the two ordered terms (q, q') and (q', q) have opposite vectors $\pm(\mathbf{b}_{qs} - \mathbf{b}_{q's})$, giving the normalized coefficient $2/(NR_s)$. The case $q = q'$, $\ell \neq \ell'$ is identical and gives the coefficient $2/(NR_t)$. Finally, when both ordered pairs are off diagonal, the four terms generated by $(q, q') \leftrightarrow (q', q)$ and $(\ell, \ell') \leftrightarrow (\ell', \ell)$ split into the two conjugate pairs $\pm\Delta_{q,q',\ell,\ell'}$ and $\pm\Delta_{q,q',\ell',\ell}$, giving the final double sum with normalized coefficient $2/(NR_s R_t)$. Thus the remaining sums run over unordered indices $q < q'$ and $\ell < \ell'$. Individual Ω -terms need not be nonnegative; the nonnegativity of the whole expression follows from its equality to the probability overlap $\sum_x \Pr_s(x) \Pr_t(x)$.

For a stratified random triple with $\nu_2((q' - q)s) = \alpha$, define the zero-factor frequency by

$$P_\alpha(q, q', s) := \frac{1}{n} \left| \left\{ 1 \leq a \leq n : \cos \left[\frac{\pi}{2} (C \Delta_{q, q', 0, 0})_a \right] = 0 \right\} \right|. \quad (\text{E32})$$

The empirical distribution of P_α is well described by

$$P_\alpha(q, q', s) \approx \mathcal{N}(\mu_\alpha, \sigma_\alpha^2), \quad \mu_\alpha = \frac{1}{4} - \frac{\lfloor \alpha/2 \rfloor - \mathbf{1}_{\alpha \text{ odd}}}{2n}, \quad \sigma_\alpha^2 = \frac{n/2 - \lfloor \alpha/2 \rfloor}{4n^2}. \quad (\text{E33})$$

The reason for this form is the staggered odd/even structure of the HP-1 matrix C . Conditioning on α freezes a trailing-zero depth in $\Delta_{q, q', 0, 0}$. Since the dyadic part of C couples odd rows to even columns, every two additional trailing zeros remove roughly one active dyadic hash constraint, giving the drift in μ_α ; the odd- α correction comes from the row-parity staggering. The remaining $n/2 - \lfloor \alpha/2 \rfloor$ active contributions behave like weakly dependent Bernoulli fluctuations, so averaging over n rows gives $\text{Var}(P_\alpha) \approx (n/2 - \lfloor \alpha/2 \rfloor)/(4n^2)$. $\square$

F. Ln-linear growth fit for the numerical Fisher-information scan

The fixed- r QFT benchmark derived above clarifies the dominant scaling mechanism, but Fig. 2 probes a slightly different quantity. There, the vertical axis is the minimum discrete Fisher information over the period window used in the numerical scan. For the finite-size numerical plot, we use the square-root boundary proxy $2 \leq r \leq \lfloor 2^{n/2} \rfloor$, matching the period scale used in the recovery experiments. The relevant QFT comparator is therefore the worst case within this dynamically scaled finite window rather than a single fixed value of r .

The leading term in Lemma 1 is monotone in R , and in the sampled regime $R \approx 2^n/r$. Consequently, the smallest QFT Fisher information inside the scan window is attained near the largest sampled period. Substituting $r_{\max}(n) = \lfloor 2^{n/2} \rfloor$ yields the window-matched estimate

$$\text{DFI}_{\text{QFT}, \min}(n) \approx \frac{4\pi^2}{9} \left[\left\lfloor \frac{2^n}{\lfloor 2^{n/2} \rfloor} \right\rfloor^2 - 1 \right] \asymp \frac{4\pi^2}{9} 2^n. \quad (\text{F1})$$

If one keeps only the dominant exponential factor, the natural-ln growth becomes

$$\ln \text{DFI}_{\text{QFT}, \min}(n) = n \ln 2 + \mathcal{O}(1). \quad (\text{F2})$$

Thus, the asymptotic QFT exponent under this square-root period scaling is $k_{\text{QFT}, \infty} = \ln 2$.

This observation suggests a simple diagnostic for the numerical data in Fig. 2. If the plotted curves retain a stable exponential dependence on the qubit count over the sampled window, then the transformed data should be well approximated by

$$\ln \text{DFI}_{\min}(n) = kn + b. \quad (\text{F3})$$

We therefore perform ordinary least-squares fits to the ln-transformed mean values of each curve shown in Fig. 2. Table I summarizes the fitted exponents; we do not repeat the transformed plot because it carries the same point set as Fig. 2.

Circuit	qubits	k	95% CI	R^2
HP-1 fixed	7-18	0.368	[0.324, 0.413]	0.972
HP-1 shared optimized	7-18	0.379	[0.354, 0.403]	0.992
HP- L random	7-18	0.398	[0.383, 0.414]	0.997
HP- L random with phase	7-18	0.352	[0.336, 0.367]	0.996

TABLE I. Effective exponents obtained from the ln-linear model $\ln \text{DFI}_{\min}(n) = kn + b$ applied to the numerical Fisher-information curves in Fig. 2.

The fitted lines account for most of the ln-scale variance across the sampled window. The coefficient of determination (R^2) ranges from 0.972 to 0.997, the ln-space root-mean-square residual lies between 0.075 and 0.217, and every fitted

slope (k) is significantly positive with a p -value $\leq 4.6 \times 10^{-9}$. In that sense, a linear regression on $\ln I_{\min}$ is justified as a finite-window descriptive model for these curves. At the same time, the fitted slope k should be interpreted as an effective exponent on the restricted range $n = 7, \dots, 18$ (where n is the number of qubits), not as a claim about the ultimate asymptotic law.

The comparison with the QFT benchmark is unambiguous. Even after matching the square-root period window used in the numerical scan, the QFT reference fit remains steeper, with $k_{\text{QFT,win}} = 0.695$ and $R^2 = 1.000$ over $n = 7, \dots, 18$. The empirical circuit families therefore grow more slowly: their fitted exponents occupy about 51%–57% of the window-matched QFT slope and of the asymptotic QFT exponent $\ln 2$. Among the plotted families, the random-layer HP circuit has the largest point estimate, near $k = 0.398$, followed by the shared optimized HP-1 circuit, near $k = 0.379$.

This comparison sharpens the interpretation of Fig. 2. The HP family circuits retain a clear exponential-in- n growth trend in the $\text{DFI}_{\min}$ over the sampled window, which is why the $\ln$ -linear fit works well, but they do not reproduce the QFT growth rate. Their advantage in the present setting is therefore not that they match the full QFT benchmark asymptotically, but that they retain a nontrivial and practically usable fraction of that information while remaining much shallower.

G. Noise-robustness experiment details

We evaluate noise robustness using the previously trained 9- and 10-qubit HP-1 plus neural-decoder pipelines from the period-recovery study. For each system size, we use a measurement budget of $1024n^2$ samples per instance, giving 82944 measurements for $n = 9$ and 102400 for $n = 10$. The candidate periods are $r \in \{9, \dots, 80\}$ for 9 qubits and $r \in \{10, \dots, 99\}$ for 10 qubits. Probability distributions are generated from the exact subgroup support of the trained HP-1 circuit, and the neural decoder checkpoint is selected using noiseless validation accuracy. The selected checkpoints occur at epoch 3 with validation top-1 accuracy 0.9959 for 9 qubits, and at epoch 10 with validation top-1 accuracy 1.0000 for 10 qubits.

The evaluation set holds out 3 shifts for each candidate period and generates 8 independent noisy redraws for each held-out shift. This yields $72 \times 3 \times 8 = 1728$ test examples for 9 qubits and $90 \times 3 \times 8 = 2160$ test examples for 10 qubits. To probe robustness after the shallow quantum transformation, we insert a single global depolarizing channel after the full HP-1 circuit and before measurement. At the probability level, this is implemented as

$$\widetilde{\text{Pr}}_{\eta}(x|r, s) = (1 - \eta)\text{Pr}(x|r, s) + \eta \frac{1}{2^n}, \quad (\text{G1})$$

where $\text{Pr}(x|r, s)$ is the ideal output distribution for period r and shift s , and η is the depolarizing strength. For each noisy distribution $\widetilde{\text{Pr}}_{\eta}$, we resample $1024n^2$ outcomes and feed the resulting bit-string batch into the fixed neural decoder. The sweep uses the values

$$\eta \in \{1, 0.464, 0.215, 0.1, 0.0464, 0.0215, 0.01, 0.00464, 0.00215, 0.001\}. \quad (\text{G2})$$

η	9-qubit accuracy	10-qubit accuracy
1.000	0.0139	0.0111
0.464	0.4913	0.8810
0.215	0.9948	1.0000
0.100	0.9954	1.0000
0.0464	0.9959	1.0000
0.0215	0.9942	1.0000
0.0100	0.9954	1.0000
0.00464	0.9959	1.0000
0.00215	0.9948	1.0000
0.00100	0.9948	1.0000

TABLE II. Held-out period-recovery accuracy under post-HP-1 global depolarizing noise.

Several features of Table II are worth noting. First, performance is essentially unchanged throughout the weak-to-moderate noise regime and remains near the noiseless baseline up to $\eta \approx 0.215$. Second, substantial degradation

appears only once the noisy distribution is strongly mixed with the uniform distribution: at $\eta = 0.464$, the accuracy drops to 0.4913 for 9 qubits and 0.8810 for 10 qubits. Finally, when $\eta = 1$, the output distribution becomes exactly uniform and is therefore independent of both the period and the shift. In that limit, all subgroup information is erased, so the best possible behavior is chance-level guessing over the candidate periods. The observed accuracies 0.0139 and 0.0111 agree exactly with the corresponding chance levels $1/72$ and $1/90$, confirming that the numerical results are consistent with the depolarizing-noise model.